Tech Eximius 2.0 · Project / Prototype Name

MediSaathi

Health help that works in airplane mode.

Reads a lab report in plain words, answers health questions in multilingual Indian languages with Gemma 4 on the phone, scans medicines, tracks vaccines and outbreaks, finds blood donors. Nothing waits for a signal.

0 bars
needed for any feature

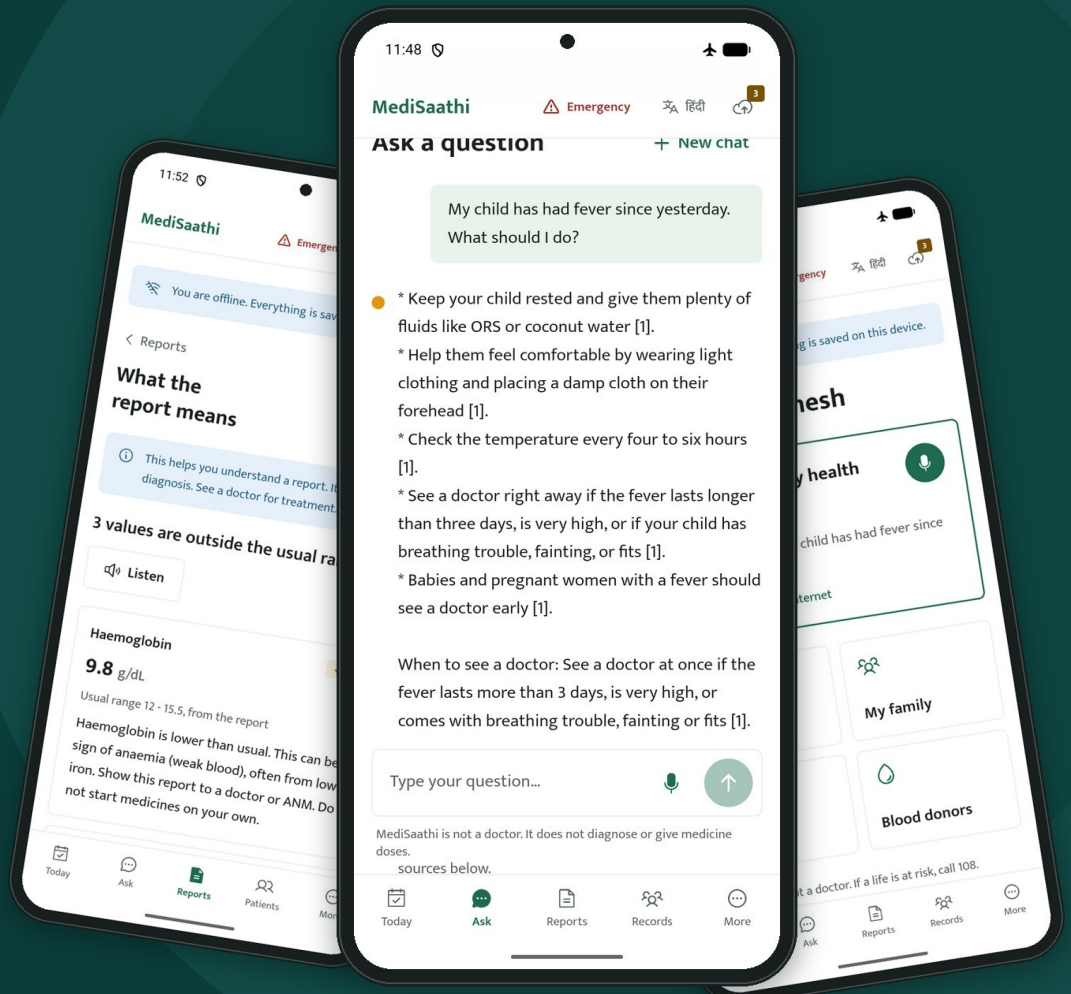
Gemma 4
runs on the phone

Multilingual
voice and text

TEAM NAME

Adamya

PS	MY	SK	SK
Parth Sharma	**Monishka Yadav**	**Shrey Kataria**	**Shresth Kataria**
Team Leader	Team Member	Team Member	Team Member



Gemma 4 answer, no internet

GitHub Repository Link: https://github.com/VenomDevX/MediSathi

Working Prototype Link: https://github.com/VenomDevX/MediSathi/raw/main/apk/MediSaathi.apk

Demo Video Link: https://youtube.com/shorts/RxIR7okYcKo

Healthcare apps fail exactly where care is needed

No signal, no help
When the network drops, cloud apps stop working in villages and relief camps.

Reports are hard to read
“Hb 9.8 g/dL” means little to most families, so worry or wrong steps follow.

Records are fragmented
Paper registers get wet or lost, and follow-up visits quietly slip.

Donor info goes stale
In an emergency, nobody knows who nearby can give blood.

	Cloud app	MediSaathi
Explain a report	✕ Online only	✓ **On device**
Patient records	✕ Online only	✓ **Local first**
Donor search	✕ Online only	✓ **Cached directory**
First-aid guidance	✕ Online only	✓ **Reviewed library**
Health answers	✕ Unverified AI	✓ **Rules + sources**

WHO WE SERVE Four users, one offline toolkit

Patient / caregiver
- Report explained simply
- Saved reports offline
- Knows when to seek care

ASHA worker
10 lakh+ in India

- Registers patients in the field
- Offline follow-up list
- Secure sync on reconnect

Relief worker
- Fast, reviewed first aid
- Cached donors and records
- Low power and storage

Supervisor / admin
- Manages staff access
- Reviews sync failures
- Runs donor consent

Four modules, one sync layer, zero bars needed

MediScan
Report OCR + plain language

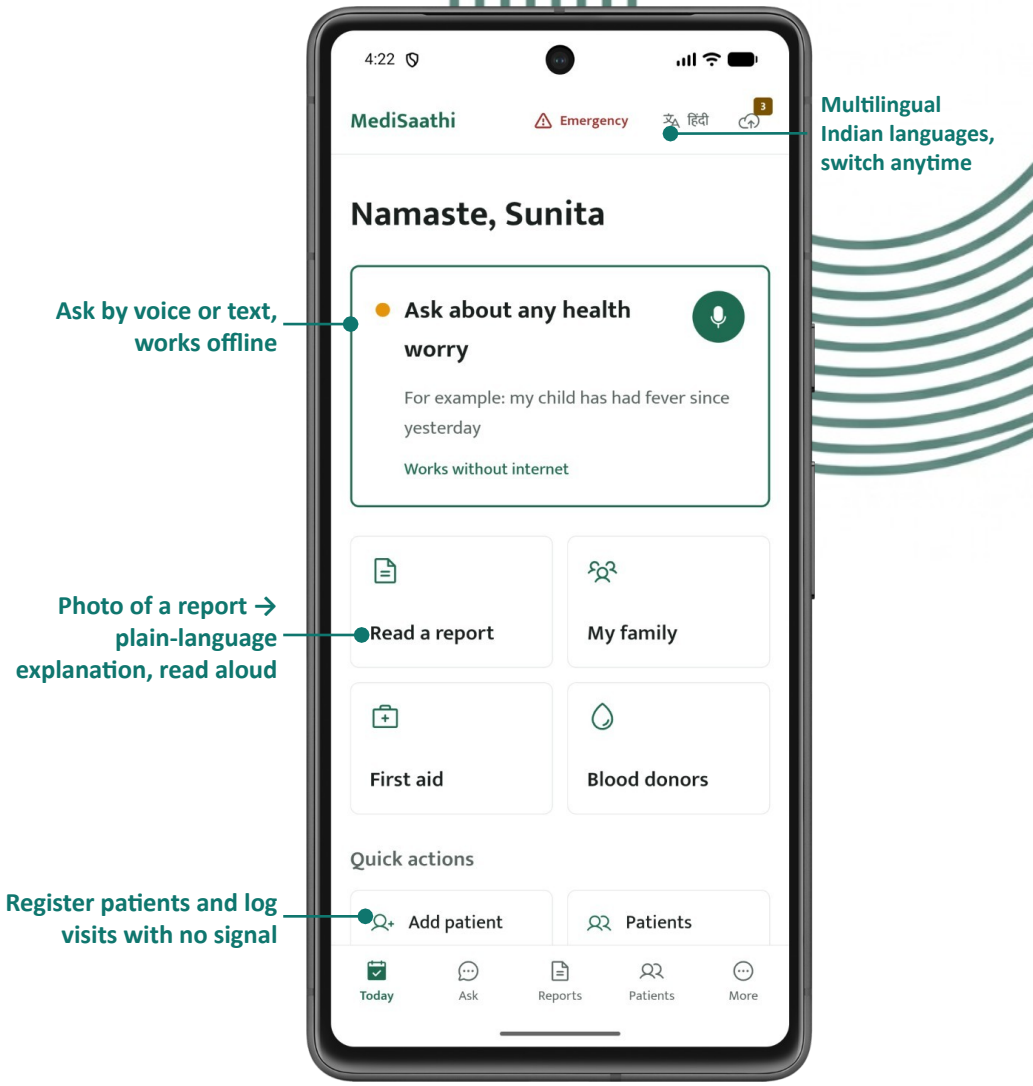
ASHA Assist
Patients, visits, follow-ups

Local DB
on device

BloodLink
Cached donor search

Health Guidance
Reviewed first-aid library

⇅ Sync service → FastAPI + PostgreSQL, when online



OUR USP Safe by design, not by prompt

Offline first
Essential workflows never wait for a server.

Privacy by default
Consent, encryption and scoped access built in.

Rules before AI
Reviewed logic decides; AI only rewords.

Humans decide
Diagnosis and treatment stay with clinicians.

Ask in your language. Gemma rewords; it never decides.

Question
Multilingual, voice or text

→ **Red flag?**

Yes → **Call 108 card**
Snake bite, not breathing, heavy bleeding: first steps, model never asked

No →

1 Library search
Reviewed first-aid cards, Hindi + English (SQLite FTS)

→ **2 Gemma 4 E2B**
On the phone via Google LiteRT-LM, uses only the matched card

→ **3 Guard**
Needs a citation, no new numbers, no medicine names

→ **4 Answer + source**
Every answer shows where it came from

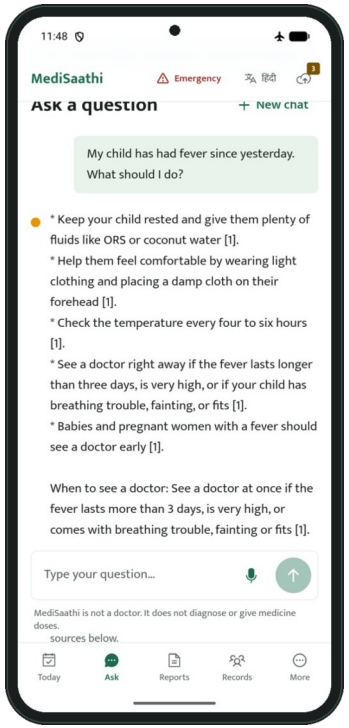
↪ Guard fails? The reviewed card is shown as is.

Safe by design: rules decide, Gemma only rewords, a guard checks every answer.

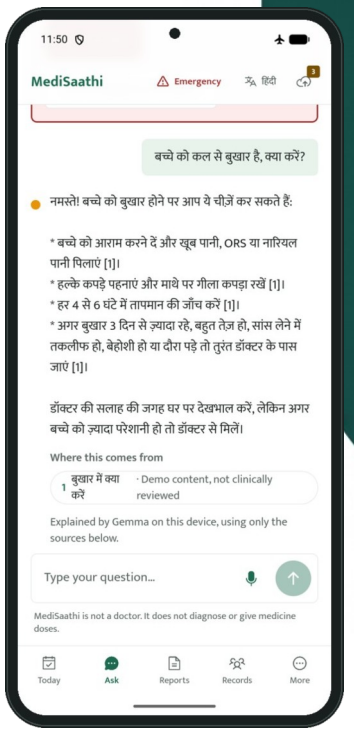
More features, all offline

- Hands-free voice Saathi
- Medicine strip scan
- Outbreak alerts on a village map
- Vaccine and pregnancy schedules
- Nearest PHC and 24-hour hospital
- SMS to blood donors
- ABHA + FHIR record export

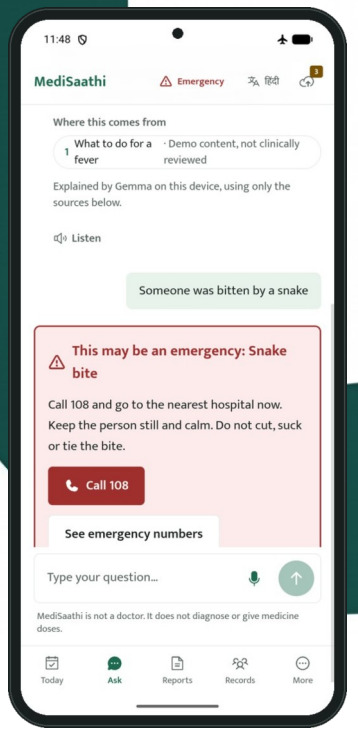
English



हिंदी



Emergency: AI skipped



108
Ambulance
Free, across India

102
Mother and child
Transport to hospital

112
National emergency
Police, fire, ambulance

Three modules working with no signal today

MODULE 01
MediScan

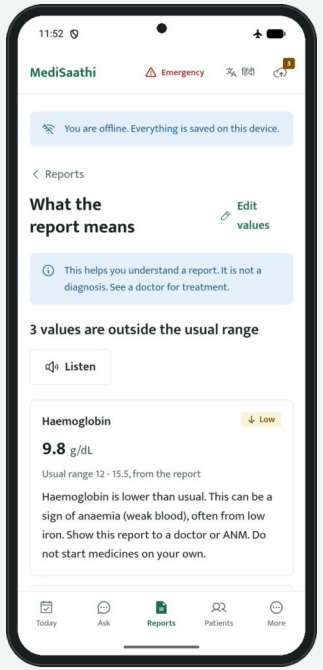


Photo or PDF of the report

Text read on the phone (Tesseract, Hindi + English)

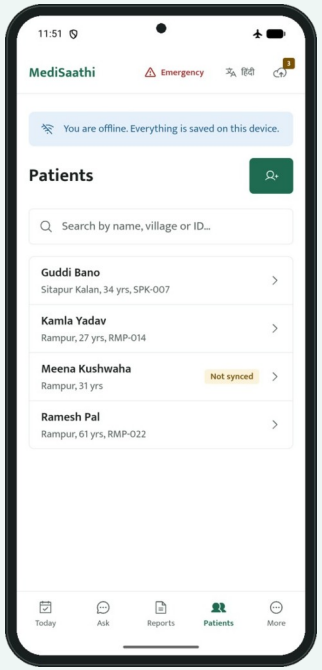
Unclear values flagged for you to fix

Range check: low, normal or high

Reviewed explanation, read aloud

Unknown tests are not guessed (“please ask your doctor”). Ranges come from the report; a fallback range is labelled. Not a diagnosis: said on every report.

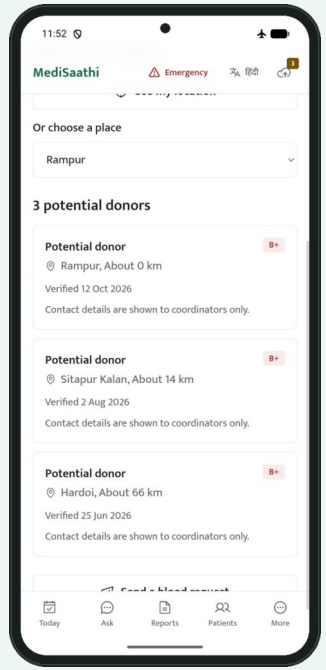
MODULE 02
ASHA Assist



- Register patients and log visits with no signal
- Everything saved on the phone first, then synced
- “Not synced” badge until the server confirms
- PIN unlock; works up to 7 days offline

Each worker only syncs their own villages; the server checks every record.

MODULE 03
BloodLink



- Same blood group only, nearest first
- Only town and distance shown, never an exact location
- “Potential donor”: the blood bank confirms
- Cached directory works offline

A donor’s number is shared only after they say yes.

One React app on web, PWA and Android, tested and ready to pilot

On the device: everything works offline

Screens	React 19, Tailwind v4, Multilingual Indian languages, voice in and out
Answers	Emergency rules → library search (SQLite FTS) → Gemma 4 → guard
Data	SQLite on the device: patients, visits, reports, sync queue
Reports	Tesseract OCR (Hindi + English) and pdf.js, on the device

Browser / PWA
Gemma via LiteRT-LM on WebGPU · sqlite-wasm on OPFS, service worker

Android (Capacitor 8)
Gemma via native LiteRT-LM, GPU then CPU · same code, same database schema

HTTPS when online →

FastAPI server

- Sign up or Google sign-in; Argon2, rotating tokens
- Idempotent push, scoped pull
- BloodLink requests with donor consent
- Optional Gemini explainer; the key stays on the server
- SQLAdmin panel, PostgreSQL in production

React + TypeScript + Vite · LiteRT-LM + Gemma 4 E2B · Capacitor 8 · FastAPI + PostgreSQL · Playwright + Vitest + pytest

Impact: built, tested and ready for a field pilot

211	**84**	**34 × 2**	**2.6 GB**	**30 s**	**Multilingual**
unit tests	backend tests	end-to-end flows	Gemma 4 E2B on Android	full Gemma answer	Indian languages
guard, sync, schedules, FHIR	sign-up, Google, scope, visit checks	on phone and laptop sizes	2.0 GB build in the browser	emulator CPU, no GPU, airplane mode	Hindi and English read aloud

Village health programmes · Disaster and relief camps · Blood-bank coordination · ABHA-linked records

Try it: Sunita (ASHA) or Ramesh (citizen), PIN 1234, offline